

Does Claude Optimize for the First Step?

Measuring Short-Horizon Preference in LLM-Guided GPU Kernel Optimization

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Kuan-Lei Wu

R14944049

GINM

r14944049@ntu.edu.tw

Sheng Chen

R11922119

CSIE

r11922119@ntu.edu.tw

Young Lee

R14942037

GICE

r14942037@ntu.edu.tw

Abstract—GPU kernel optimization is path-dependent: a first edit can produce a quick speedup and then plateau, while a weaker first edit may open a better trajectory. This report studies whether Claude’s first-move preference reflects immediate speedup or the best speedup found after continuing optimization. For four HecBench CUDA kernels, Claude generated and ranked five candidate optimization plans under either a default prompt or an explicit long-horizon prompt. Each plan was executed independently by a Worker branch for a fixed budget of seven successful correctness-checked iterations. Using a geometric-normalized rank-step penalty, the default prompt showed a small iteration-1 penalty of 3.0% per lower agent-ranking position but essentially no final-speedup penalty, 0.0%. The long-horizon prompt kept similar iteration-1 alignment, 2.9%, but raised final-speedup alignment to 11.2%. It also reduced top-choice final-speedup regret from 26.1% to 10.8%. The logs show why prompting helps but is insufficient: in these runs, Claude still shows salient-metric fixation, wrong mechanism selection, algorithmic elegance bias, and resource-tradeoff misses.

Index Terms—GPU optimization, CUDA, LLM agents, prompt engineering, performance analysis, rank-step penalty

Project Category: ☐ A (Self-chosen topic) ☒ B (AI-agent code optimization)
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I. INTRODUCTION

Large language model (LLM) coding agents are increasingly used as practical optimization assistants. A simple workflow is attractive: give the agent a kernel, ask it to optimize, run the benchmark, and repeat. The risk is that kernel optimization is not a sequence of independent local edits. A CUDA change that improves the first run can consume registers, reduce occupancy, add shared-memory pressure, or fix only an easy local symptom. Another direction may start slowly but create more room for later changes.

This report asks whether Claude’s first-move preference is aligned with short-term payoff, longer-horizon speedup, or neither. To measure that preference, Claude is asked to generate and rank five candidate optimization plans. This generate-and-rank step is a measurement probe rather than a normal user interface: it exposes which starting directions Claude prefers before any implementation outcomes are known. Each plan is then run independently by a Worker branch under the same correctness and iteration budget.

The experiment also asks whether this preference changes when trajectory value is stated explicitly. In a default prompt, Claude ranks plans by what it would try first. In a long-horizon prompt, Claude is told that each plan will be executed by a separate Worker for a fixed budget and is asked to rank plans by which starting direction is most likely to reach the highest speedup within that budget. The target is not the number seven itself; seven successful iterations are simply the controlled search budget used to approximate continuing optimization.

The contribution is not a new CUDA optimization technique. The contribution is a controlled analysis of LLM-guided optimization behavior. We compare the two prompt conditions, use magnitude-aware metrics instead of pure rank agreement, and connect the numerical results to Worker logs. The main finding is that Claude’s default ranking weakly predicts which plan works immediately, but almost does not predict which direction reaches the best speedup after the optimization budget. Long-horizon prompting improves this alignment and reduces regret, but the logs show persistent causal reasoning errors over profiler signals and GPU resource tradeoffs.

II. BACKGROUND

A. Kernels and Optimization Angles

The experiment uses CUDA kernels from HecBench, a heterogeneous-computing benchmark suite with GPU implementations of common algorithms and application kernels [1]. We use four kernels whose optimization behavior is understandable from code and profiler evidence, while still offering multiple plausible optimization angles. This selection supports interpretable trajectory analysis rather than broad benchmark coverage.

B. Why First Direction Matters

CUDA performance constraints are coupled. Shared memory may reduce global traffic but lower occupancy. Register-heavy rewrites may remove memory operations but cause spilling. Changing precision can reduce bandwidth pressure and also enable larger tiles. Grid size can reduce launch or pass overhead but increase atomic contention. These tradeoffs

TABLE I
KERNELS AND OPTIMIZATION ANGLES.

Kernel	Main tension	Plausible angles
bitonic-sort	Mapping a sorting network onto GPU threads.	Shared memory, global-memory layout, divergence, launch structure.
histogram	Parallelism versus atomic contention.	Grid shape, pass fusion, allocation traffic, memory layout.
softmax	Memory latency versus memory traffic.	Read reduction, occupancy, block size, latency hiding.
stencil3d	Bandwidth, precision, and tiling interact.	Precision, coalescing, block shape, shared-memory tiling.

mean the first edit is not merely a local patch; it selects a region of the optimization landscape.

This coupling makes LLM-guided optimization interesting. An agent can know many plausible CUDA tactics, but the valuable behavior is choosing a first direction whose later refinements have headroom. The paper therefore measures both iteration-1 speedup and the best speedup found after a fixed optimization budget. The first is the payoff a naive loop sees immediately. The second is closer to the goal of this line of work: finding a strong kernel implementation.

III. METHODOLOGY

A. Agent Ranking Probe and Worker Execution

For each kernel and prompt condition, Claude generated five candidate optimization plans and ordered them from P1 to P5. We call this order the *agent ranking*. This ranking is not intended to model a normal user interface; it is a probe for Claude’s preference before outcome feedback. Each ranked plan was assigned to an independent Worker branch. The Worker received only that plan, edited an allowlisted source-file set, and repeatedly ran `./src/run.sh`. A successful iteration means the run finished and `out/latest_run.json` reported `status: correct`. Compile errors, correctness failures, profiling infrastructure errors, and iteration-limit events were logged but did not count toward the seven successful iterations.

This separation is important. If one Claude session repeatedly optimized a kernel, later decisions would be affected by earlier outcomes. Here, the generate-and-rank step is measured before any Worker sees a result. The five branches start from the same baseline and are evaluated under the same rules, so differences between branches can be interpreted as differences between starting directions and the Worker trajectories they enable.

B. Prompt Conditions

Condition A used a default optimization prompt. Claude was asked to propose exactly five distinct plans and rank them from highest to lowest priority based on what it would try first to improve performance. The prompt did not mention downstream Workers, the seven-iteration budget, or trajectory evaluation.

Condition B used a long-horizon prompt. Claude was told that a separate Worker would execute each ranked plan for

exactly seven iterations. It was asked to rank plans by which starting direction was most likely to achieve the highest peak speedup at any point within those iterations, considering how the first direction opens or blocks later optimizations. Both conditions used the same kernel sources, benchmark harness, correctness rule, Worker loop, and speedup measurement. The intended independent variable is the horizon framing of the plan ranking.

C. Magnitude-Aware Metrics

Pure rank metrics are too coarse for this question because they treat near ties and large losses equally. We instead use a magnitude-aware rank-step penalty. For one fixed kernel, prompt condition, and evaluation horizon, let s_p be the speedup of the plan at ranking position $p \in \{1, \dots, 5\}$. We first divide each speedup by the geometric mean of the five plans:

$$g = \left(\prod_{p=1}^5 s_p \right)^{1/5}, \quad y_p = \frac{s_p}{g}. \quad (1)$$

Thus $y_p = 1$ means average for that kernel, $y_p = 1.10$ means 10% above that kernel’s typical plan, and $y_p = 0.90$ means 10% below it. After this normalization, we pool the points for a condition and horizon and fit a straight line from P1 to P5:

$$y_p = a + bp + \epsilon_p. \quad (2)$$

The reported rank-step penalty is $-100b$. If the penalty is 3%, then moving one position lower in Claude’s ranking corresponds to about 3% lower speedup relative to that kernel’s typical branch. Positive means the ranking predicts speedup; zero means it does not; negative means it is inverted.

We use this penalty at two horizons: iteration 1 and best speedup after the fixed Worker budget. The *greediness score* is the iteration-1 penalty minus the final-speedup penalty. Positive greediness means the ranking is more aligned with immediate gain; negative greediness means it is more aligned with longer-horizon gain. Finally, we report top-choice regret:

$$R = \frac{s_{\text{best}} - s_1}{s_{\text{best}}}. \quad (3)$$

This regret metric estimates the cost of following Claude’s first-ranked plan instead of exploring all five starting directions. It also makes a one-rank miss on a near tie count less than a one-rank miss that leaves a large amount of speedup on the table.

IV. EXPERIMENTAL SETUP

The design creates 40 Worker branches: four kernels, two prompt conditions, and five ranked plans per condition. All experiments were conducted using Claude Code (Sonnet 4.6, Medium Effort) as the underlying agent model. Each branch starts from the same original kernel baseline for its kernel. Speedup is measured against that fixed baseline, and correctness is required before timing contributes to analysis. The recorded experiment environment was an NVIDIA GeForce RTX 3080 Ti with NVIDIA driver 580.126.09, CUDA driver

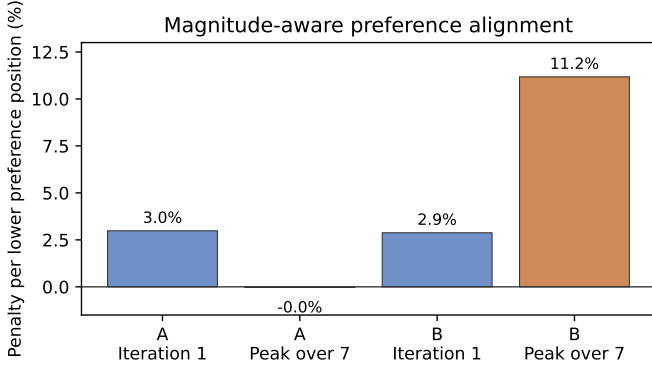


Fig. 1. Rank-step penalty by prompt condition and horizon. Higher final penalty means Claude’s ranking better predicts which starting direction reaches the best speedup after the fixed optimization budget.

TABLE II
MAGNITUDE-AWARE ALIGNMENT OF AGENT RANKING WITH SPEEDUP.
PENALTY IS PERCENT LOWER SPEEDUP PER ONE-STEP LOWER RANKING POSITION.

Condition	Iter. 1	Final	Greedy
A default	3.0%	0.0%	+3.0 pp
B long-horizon	2.9%	11.2%	-8.3 pp

version 13.0, and nvcc 12.0.140. Each timed run used one warmup execution followed by five measured executions, with the median time used for speedup. Branches were run sequentially, and the execution plan required an idle-GPU check before launching each Worker branch. Across all conditions, 280 of 308 Worker attempts produced correct outputs. The non-counted attempts were 9 compile errors, 8 correctness failures, 3 profiling infrastructure errors, and 8 iteration-limit events. The median timing relative standard deviation across counted runs was 0.108%. All report artifacts are regenerated by:

```
python3 analysis/scripts/compute_metrics.py
```

V. RESULTS

A. Ranking Alignment by Horizon

Figure 1 and Table II show the central result. Under the default prompt, each one-step lower agent-ranking position corresponds to 3.0% lower iteration-1 speedup, but essentially no loss in the best speedup reached after the Worker budget. Thus the default ranking is mildly useful for immediate payoff but not for final trajectory value.

The long-horizon prompt changes what the ranking predicts. Its iteration-1 penalty remains similar at 2.9%, but its final-speedup penalty rises to 11.2% per lower ranking position. The greediness score flips from +3.0 percentage points to -8.3 percentage points: the ranking becomes more aligned with final speedup than with first-step payoff. This is the cleanest evidence that the prompt does not simply make every branch faster. It changes Claude’s preference over starting directions. The per-kernel penalties support the same direction

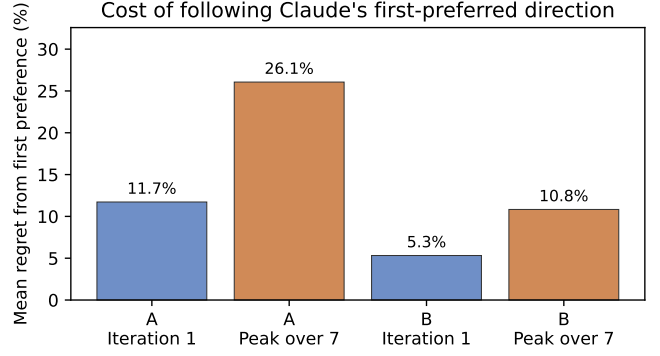


Fig. 2. Top-choice regret from following P1 instead of the best of P1–P5. The long-horizon prompt reduces regret at both horizons, especially for the final speedup found after the fixed Worker budget.

TABLE III
TOP-CHOICE REGRET: SPEEDUP LEFT ON THE TABLE BY FOLLOWING P1 INSTEAD OF THE BEST OF P1–P5.

Condition	Iter. 1 regret	Final regret
A default	11.7%	26.1%
B long-horizon	5.3%	10.8%

but also show heterogeneity: Condition B improves final-speedup alignment for bitonic-sort, histogram, and stencil3d, while softmax remains slightly inverted. Thus the aggregate result is not a universal success claim; it is a strong pattern with one clear exception.

B. Practical Regret from Trusting P1

Figure 2 and Table III give the practical version of the result. If a user follows Claude’s first-ranked plan under the default prompt, the average final-speedup regret is 26.1%. With long-horizon prompting, this drops to 10.8%. The risk remains, but the cost of trusting the first direction is substantially smaller.

The per-kernel regret values clarify where the average comes from. In the default condition, P1 leaves 41.6% final-speedup regret on bitonic-sort, 36.2% on histogram, and 25.1% on stencil3d, while softmax has only 1.5% regret. Therefore the default prompt’s cost is not uniform. Its main failure is that it sometimes misses the large-headroom direction, and pure ordering metrics would not distinguish those costly misses from the small softmax near tie.

C. Raw Outcome Pattern

Table IV shows the best speedup found by each ranked plan. It makes the aggregate metrics inspectable: Condition A never has P1 as the final-speedup winner, while Condition B has P1 winners for bitonic-sort and stencil3d. Histogram B remains imperfect because P2 wins strongly. Softmax is the important magnitude lesson: P5 wins in both conditions, but the performance cluster is comparatively tight, especially in Condition A.

TABLE IV
BEST SPEEDUP WITHIN THE FIXED WORKER BUDGET BY AGENT-RANKING POSITION. P1 IS CLAUDE’S FIRST-RANKED PLAN.

Kernel	Cond.	P1	P2	P3	P4	P5	Winner
bitonic-sort	A	1.382	1.409	2.366	1.001	1.000	P3
histogram	A	0.971	1.521	1.014	1.052	1.519	P2
softmax	A	1.653	1.620	1.656	1.619	1.677	P5
stencil3d	A	2.198	2.204	2.932	2.751	2.318	P3
bitonic-sort	B	1.915	1.449	1.529	1.702	1.118	P1
histogram	B	1.464	2.286	1.002	1.002	1.025	P2
softmax	B	1.438	1.512	1.482	1.513	1.553	P5
stencil3d	B	4.985	2.296	2.298	1.476	2.778	P1

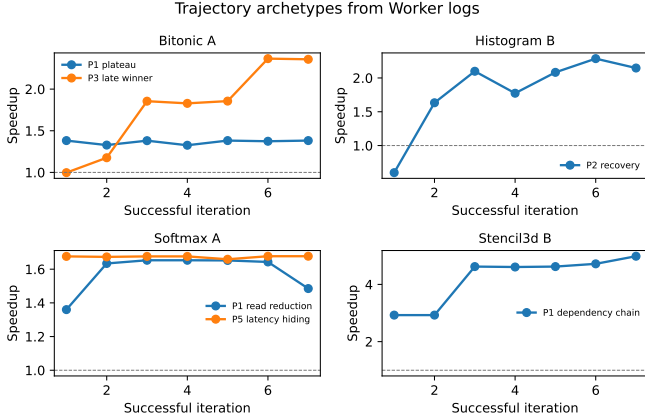


Fig. 3. Representative Worker trajectories. First-iteration speedup can overstate or understate the value of the starting direction.

D. Trajectory Evidence

Figure 3 shows representative trajectories. Bitonic-sort A shows an immediate-win plateau: P1 starts and ends at $1.382\times$, while P3 starts near baseline and later reaches $2.366\times$. Histogram B shows the opposite: P2 starts at $0.599\times$ because the first implementation has too few blocks, but recovers to $2.286\times$ after grid tuning. Stencil3d B shows the positive case for long-horizon prompting, where P1 moves from $2.927\times$ to $4.623\times$ to $4.985\times$ through a coherent dependency chain.

E. Headroom After the First Step

Table V summarizes the same point numerically. Histogram B P2 is the most extreme case: the first successful implementation is only $0.599\times$, but the branch later reaches $2.286\times$, a $3.82\times$ recovery. Bitonic A P3 similarly starts near baseline and later wins. In contrast, Bitonic A P1 has no headroom after its quick first win. These cases show why evaluating only the next benchmark run would reward the wrong behavior for this task.

VI. DISCUSSION AND ROOT-CAUSE ANALYSIS

A. Failures Are Structured

The quantitative pattern is not random. Table VI summarizes the main log-derived cases. These cases explain why the default ranking can look reasonable in the first iteration while failing to identify the best trajectory. They also show why

TABLE V
HEADROOM AFTER THE FIRST SUCCESSFUL ITERATION.

Case	P	First	Peak	Recovery
Histogram B	P2	0.599	2.286	$3.82\times$
Bitonic A	P3	0.997	2.366	$2.37\times$
Histogram A	P5	0.656	1.519	$2.32\times$
Stencil3d B	P1	2.927	4.985	$1.70\times$
Bitonic A	P1	1.382	1.382	$1.00\times$

the long-horizon prompt helps only sometimes: Claude must still choose the correct causal mechanism. Here, salient-metric fixation means over-weighting a visible profiler symptom, algorithmic elegance bias means favoring a theoretically appealing rewrite over a simpler hardware-effective change, and a resource-tradeoff miss means improving one GPU resource while overlooking the cost paid in another.

In bitonic-sort A, P1 fused shared-memory stages and immediately achieved $1.382\times$, but logs reported significant shared-memory bank conflicts and increased registers per thread. The branch never improved beyond that. P3 began with a host-side cleanup that did almost nothing, $0.997\times$, but later found the real structural path: shared-memory efficiency, linear thread-to-comparison mapping, and reduced divergence. It peaked at $2.366\times$. This is the clearest example of why first-step speedup is a weak proxy for trajectory value: the quick win addressed visible memory behavior but left little headroom.

Histogram B shows the complementary risk. P2 initially slowed to $0.599\times$ because the first single-pass implementation used too few blocks for a 1920-by-1080 image. The strategy itself was strong: it removed a second kernel launch and a global write/read round trip. After grid sizing restored parallelism and balanced atomic contention, the branch reached $2.286\times$. A strict “must improve immediately” policy would have discarded one of the best directions. This is directly relevant to agent search design: early failure can be an implementation parameter error rather than a bad hypothesis.

B. Profiling-Aware Is Not Causal

Softmax is the key reasoning-quality counterexample. In Condition A, Claude noticed memory behavior and ranked read-reduction plans highly. P1 reduced global reads from three passes to one and halved exponential calls, but occupancy fell from 74.1% to 30.7%. P5, a lower-ranked block-per-slice direction, won by increasing per-slice parallelism: occupancy

TABLE VI
ROOT-CAUSE TAXONOMY FROM WORKER LOGS.

Case	Trajectory	Reasoning category	Lesson
Bitonic A	Immediate plateau vs. slow-start winner	Under-valued structural headroom	First-step speedup can hide limited headroom.
Histogram B	Bad first implementation recovers	Productive recovery	Do not prune solely on the first attempt.
Softmax A/B	Close outcomes, wrong mechanism	Correct bottleneck but wrong lever; elegance bias	Need causal mechanism selection, not only profiler awareness.
Stencil3d B	Dependency-chain success	Correct direction and mechanism	Long-horizon prompting helps when later moves build on the first.

rose to 94.7% and memory stalls fell from 94.5% to 60.1%. The practical regret was small in Softmax A because outcomes were close, but the mechanism lesson is important: the stronger lever was latency hiding, not simply reducing memory traffic.

Condition B backfired differently. Claude ranked online softmax first, an algorithmically elegant rewrite that reduced a global read pass. It peaked at $1.438\times$. The fifth-ranked branch, simple block-size tuning, reached $1.553\times$ by improving scheduler utilization. The long-horizon prompt encouraged a sophisticated algorithmic story, but the hardware bottleneck favored a simpler launch-configuration knob. Thus access to profiling evidence is not enough; the agent needs causal resource accounting: which metric is the bottleneck, which is a symptom, and what cost another resource pays.

C. When Long-Horizon Prompting Helps

Stencil3d B shows the success case. Claude ranked double-to-float first. The first step halved memory traffic and raised occupancy, reaching $2.927\times$. The Worker then used profiler evidence to adjust block shape, raising global load efficiency and speedup to $4.623\times$, and later tuned XTILE to reach $4.985\times$. The first move was valuable because it unlocked later coalescing and tiling improvements. This suggests that long-horizon prompting helps most when the dependency chain between first edit and later optimizations is visible.

The contrast with softmax is important. Both kernels are memory-related, but only stencil3d offered an obvious chain where one foundational resource change made later optimization easier. In softmax, several plausible mechanisms competed: fewer reads, more occupancy, more outstanding work, and launch configuration. The long-horizon prompt could not by itself decide which mechanism had causal leverage. This points to a design implication: prompting should be paired with tools that force the agent to maintain a bottleneck hypothesis ledger and account for resource tradeoffs.

D. Implications for Agent Design

The results suggest three concrete requirements for LLM-based kernel optimization systems. First, the system should preserve multiple hypotheses instead of committing to the first fast improvement. Histogram B P2 would be lost under a strict first-step-improvement filter, even though it becomes the best histogram trajectory. Second, the system should evaluate magnitude, not only order. Softmax A is an ordering miss, but its regret is small; bitonic A and histogram A are the cases where the first choice is actually costly. Third, the agent

needs structured causal accounting over GPU resources. A useful debugging trace should record the current bottleneck hypothesis, the expected resource tradeoff, the observed profiler change, and whether the evidence supports or contradicts the hypothesis. That is the missing layer between having profiler data and using it correctly.

E. Threats to Validity

The study is small: four kernels and five generated plans per condition. The statistics are descriptive rather than significance claims, and the findings should be read as evidence about this controlled workflow rather than a broad claim about all kernels or all LLM agents. Worker quality also affects measured trajectory value; a branch may underperform because the Worker did not find the best implementation of its assigned plan within the budget. This is partly intentional, because the experiment evaluates a realistic plan-generation plus Worker workflow rather than oracle implementations. Finally, results may vary with GPU architecture, prompt wording, and benchmark choice. The value of the study is as a controlled diagnostic of Claude’s optimization preference, not as a broad claim about all kernels or all LLMs.

VII. CONCLUSION

Claude’s default first-move preference is mildly aligned with immediate speedup but weakly aligned with the best speedup found after continuing optimization. Explicit long-horizon prompting changes this behavior: the ranking becomes more predictive of final speedup and top-choice regret falls from 26.1% to 10.8%. Prompting alone is not enough. The logs reveal recurring causal failures around salient metrics, mechanism selection, algorithmic elegance, and resource tradeoffs. The practical lesson is to treat Claude’s first plan as a hypothesis, not a decision: robust LLM optimization systems should preserve multiple hypotheses, evaluate trajectory value, and provide structured tools for causal GPU-resource accounting.

REFERENCES

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- [2] NVIDIA, “CUDA C++ Programming Guide.” [Online]. Available: <https://docs.nvidia.com/cuda/cuda-c-programming-guide/>
- [3] NVIDIA, “Nsight Compute User Guide.” [Online]. Available: <https://docs.nvidia.com/nsight-compute/>
- [4] Anthropic, “Claude Code.” [Online]. Available: <https://www.anthropic.com/claude-code>

APPENDIX A INPUT PROMPTS

The human prompt to the plan-generating Claude session was intentionally short; the generated condition README contained the full task, source code, profiling digest, and output contract.

Read README.md and write response.json exactly as instructed.

The human prompt to each Worker branch was:

Read README.md and strictly follow it.

Full prompt surfaces are documented in:

```
README.md
EXPERIMENT_PLAN.md
experiments/<kernel>/cond_<A|B>/README.md
experiments/<kernel>/cond_<A|B>/workers/branch_<rank>/
  README.md
```

APPENDIX B SYSTEM PROMPT AND AGENT INSTRUCTIONS

The experiment did not rely on a separate hidden prompt authored during the experiment. The controlled agent instruction surface was the generated Planner and Worker README text. The role instructions embedded in those files are excerpted below.

Planner role instruction:

You are a CUDA optimization expert.

Worker role instruction:

You are the Worker agent. Optimize this kernel according to the assigned plan below.

The Worker README also instructed the agent to stay in its branch directory, edit only allowlisted source files, run `./src/run.sh` after each edit, treat failed attempts as non-counting, and maintain `LOG.md`.

APPENDIX C PROMPT CONDITION EXCERPTS

Condition A opening:

```
# CUDA Kernel Optimization Request: <kernel>

You are a CUDA optimization expert. The user wants to
optimize the CUDA kernel below.

Propose exactly five distinct optimization plans and rank
them from highest to lowest priority based on what you
would try first to improve this kernel's performance.
```

Condition B opening:

```
# CUDA Kernel Optimization Request: <kernel>

You are a CUDA optimization expert. A separate Claude Code
Worker will execute each ranked plan for exactly 7
iterations.

Propose exactly five distinct optimization plans and rank
them by which starting direction is most likely to
achieve the highest peak speedup at any point within
those 7 iterations. Consider how the first direction
opens or blocks later optimizations.
```

Shared output contract:

Write your answer to exactly this file:

response.json

```
response.json must be a JSON array of exactly five objects.
Each object may contain only these two fields:
- rank
- plan
```

APPENDIX D REPRESENTATIVE AGENT OUTPUT

Softmax Condition B P1, representative plan:

```
{
  "rank": 1,
  "plan": "Implement online (single-pass) softmax to
eliminate one full global read pass, combined with
float4 vectorized loads. Currently softmax2 makes 3
separate warp-strided passes over global memory ...
Target bottleneck: the 3x global memory read passes
... reducing to 2 passes ... Correctness risk: the
online rescaling formula ..."
}
```

APPENDIX E CHANGES MADE TO THE AGENT WORKFLOW

The experiment used a generated Planner/Worker harness rather than a single continuous agent session. The key workflow changes were:

- Planner/Worker split, so plan preference is measured before implementation outcomes are known.
- Two plan-generation prompts: default priority ordering and explicit long-horizon trajectory ordering.
- Five isolated Worker branches per condition.
- Seven-successful-iteration rule; failed attempts are logged but do not count toward the budget.
- Correctness gate before timing contributes to analysis.
- Fixed JSON contracts for `response.json`, `per-run.run.json`, and `condition-level.analysis.json`.
- Worker source-edit allowlist and protected-file checks.
- Reproducible artifact generation through the report artifact script.

Architecture and file-format details are documented in:

```
README.md
EXPERIMENT_PLAN.md
```

APPENDIX F ADDITIONAL RESULTS TABLES

TABLE VII
PER-KERNEL FINAL-SPEEDUP REGRET FROM FOLLOWING P1.

Kernel	Cond.	Winner	P1	Regret
bitonic	A	P3	1.382	41.6%
histogram	A	P2	0.971	36.2%
softmax	A	P5	1.653	1.5%
stencil3d	A	P3	2.198	25.1%
bitonic	B	P1	1.915	0.0%
histogram	B	P2	1.464	35.9%
softmax	B	P5	1.438	7.4%
stencil3d	B	P1	4.985	0.0%

APPENDIX G REPRESENTATIVE LOG EVIDENCE

TABLE VIII
SHORT LOG EXCERPTS USED FOR ROOT-CAUSE ANALYSIS.

Claim	Source	Excerpt
Bitonic A P1 plateau	.../bitonic-sort-cuda/cond_A/branch_1/LOG.md	"Fusing smem stages gives 1.38x speedup... significant smem bank conflicts... Registers/thread increased to 39."
Bitonic A P3 recovery	.../bitonic-sort-cuda/cond_A/branch_3/LOG.md	"Global load efficiency improved to 100%, occupancy to 92.4%. Shared memory approach is the right direction."
Histogram B P2 first failure	.../histogram-cuda/cond_B/branch_2/LOG.md	"64 blocks is not enough to cover the 1920x1080 image efficiently... serializing work."
Softmax A mechanism miss	.../softmax-cuda/cond_A/branch_1/LOG.md	"Smem reduced global loads from 3x to 1x and halved expf, but occupancy dropped from 74.1% to 30.7%."
Softmax B simple winner	.../softmax-cuda/cond_B/branch_5/LOG.md	"BLOCK_SIZE=512 improved from 1.48x to 1.55x. More warps per SM improves scheduler utilization."
Stencil3d B chain	.../stencil3d-cuda/cond_B/branch_1/LOG.md	"Float halved memory traffic and boosted occupancy from 79.6% to 96.7%."

Complete evidence indices are included in:

```
analysis/evidence/case_evidence_index.md
analysis/evidence/log_taxonomy.csv
```

APPENDIX H ANALYSIS ARTIFACTS

The final report numbers come from:

```
analysis/data/metric_summary.csv
analysis/data/greediness_summary.csv
analysis/data/top_choice_regret_summary.csv
analysis/data/top_choice_regret_by_kernel.csv
analysis/data/peak_speedup_by_preference.csv
analysis/data/headroom_recovery.csv
analysis/notes/results_summary.md
```